

11th grade
Learning evidence T3 Statistical hypothesis testing
supplementary material solutions

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C1 Describes the concept of statistical hypothesis.

Problem 1 — School Water Bottle Capacity

Problem. A school supplier claims that its reusable water bottles hold an average of 750 mL. A quality team measures a sample of $n = 36$ bottles and wants to evaluate the claim.

- Population parameter: mean capacity μ (mL).
- Significance level: $\alpha = 0,05$.

Question. Explain what a statistical hypothesis is in this context, and set up a hypothesis test to check the supplier's claim.

Solution.

A statistical hypothesis is a statement about a population parameter (here, the true mean capacity μ). We use sample data to decide whether the claim is reasonable.

Hypotheses.

$$H_0 : \mu = 750 \text{ mL}$$

$$H_a : \mu \neq 750 \text{ mL}$$

Type of test. Two-tailed, because the concern is any difference from 750 mL.

Test statistic. If the population standard deviation is known or a large sample is used, use

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Reference distribution. Standard normal under H_0 .

Decision rule. Compare the P-value to $\alpha = 0,05$ (or use the two-tailed critical region).

Conclusion in context. Decide whether the sample provides evidence that the average capacity differs from 750 mL.

Problem 2 — Library Study Time Claim

Problem. A school bulletin states that students study an average of 2,5 hours per day in the library. A survey of $n = 40$ students is conducted to check the statement.

- Population parameter: mean study time μ (hours).
- Significance level: $\alpha = 0,10$.

Question. Describe the statistical hypothesis in this setting and state the hypotheses to be tested.

Solution.

The statistical hypothesis concerns the population mean study time μ . The claim $\mu = 2,5$ hours is tested using sample data.

Hypotheses.

$$H_0 : \mu = 2,5 \text{ hours}$$

$$H_a : \mu \neq 2,5 \text{ hours}$$

Type of test. Two-tailed because the claim is an equality and deviations in either direction matter.

Test statistic and reference distribution. Use

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}},$$

with a standard normal reference distribution (or a t distribution if σ is unknown).

Decision rule. Compare the P-value to $\alpha = 0,10$.

Conclusion in context. Decide if the data support or contradict the bulletin's mean study time.

C2 Identifies the null and alternative hypothesis in a statistical test.

Problem 1 — Bakery Cookie Weight

Problem. A bakery advertises that its cookies weigh 35 g on average. A sample of $n = 25$ cookies has mean $\bar{x} = 33,8$ g, and the population standard deviation is known to be $\sigma = 3,0$ g.

- Population parameter: mean cookie weight μ (g).
- Significance level: $\alpha = 0,05$.

Question. Identify the null and alternative hypotheses and perform the test.

Solution.

Hypotheses.

$$H_0 : \mu = 35 \text{ g}$$

$$H_a : \mu < 35 \text{ g}$$

Type of test. One-tailed (left) because the concern is underfilling.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{33,8 - 35}{3,0/\sqrt{25}} = \frac{-1,2}{0,6} = -2,00.$$

Reference distribution. $Z \sim N(0, 1)$ under H_0 .

P-value. $P(Z \leq -2,00) = 0,0228$.

Decision. Since $0,0228 < 0,05$, reject H_0 .

Conclusion. There is evidence that the mean cookie weight is less than 35 g.

Problem 2 — Campus Wi-Fi Reliability

Problem. The IT office claims that campus Wi-Fi is available at least 95 % of the time. In a random sample of $n = 200$ checks, the network is available in 186 checks.

- Population parameter: availability proportion p .
- Significance level: $\alpha = 0,05$.

Question. Identify the null and alternative hypotheses and test the claim.

Solution.

Hypotheses.

$$H_0 : p = 0,95$$

$$H_a : p < 0,95$$

Type of test. One-tailed (left) because the claim is “at least 95 %,” and we test for a decrease.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,93 - 0,95}{\sqrt{0,95(0,05)/200}} \approx \frac{-0,02}{0,0154} \approx -1,30.$$

Reference distribution. Standard normal under H_0 .

P-value. $P(Z \leq -1,30) \approx 0,0968$.

Decision. Since $0,0968 > 0,05$, fail to reject H_0 .

Conclusion. The sample does not provide enough evidence that availability is below 95 %.

C3 Determines whether a test should be one-tailed or two-tailed.

Problem 1 — Bus Arrival Time

Problem. A city transport report states that buses arrive on average in 12 minutes. Students suspect that buses take longer. A sample of $n = 50$ arrivals shows $\bar{x} = 12,8$ minutes with $\sigma = 2,5$ minutes.

- Population parameter: mean arrival time μ (minutes).
- Significance level: $\alpha = 0,05$.

Question. Decide whether the test is one-tailed or two-tailed, then perform the test.

Solution.

Hypotheses.

$$H_0 : \mu = 12$$

$$H_a : \mu > 12$$

Type of test. One-tailed (right) because the concern is specifically “longer than 12 minutes.”

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{12,8 - 12}{2,5/\sqrt{50}} \approx \frac{0,8}{0,3536} \approx 2,26.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $P(Z \geq 2,26) \approx 0,0119$.

Decision. Since $0,0119 < 0,05$, reject H_0 .

Conclusion. There is evidence that average bus arrival times exceed 12 minutes.

Problem 2 — Sports Drink Sugar Content

Problem. A sports drink label states that the sugar content is 12 g per serving. A lab checks if the actual mean differs from the label. A sample of $n = 30$ shows $\bar{x} = 11,5$ g with $\sigma = 1,8$ g.

- Population parameter: mean sugar content μ (g).
- Significance level: $\alpha = 0,01$.

Question. Decide whether the test should be one-tailed or two-tailed, then perform the test.

Solution.

Hypotheses.

$$H_0 : \mu = 12$$

$$H_a : \mu \neq 12$$

Type of test. Two-tailed because the goal is to check for any difference from the label.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{11,5 - 12}{1,8/\sqrt{30}} \approx \frac{-0,5}{0,3286} \approx -1,52.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $2P(Z \leq -1,52) \approx 2(0,0643) = 0,1286$.

Decision. Since $0,1286 > 0,01$, fail to reject H_0 .

Conclusion. There is insufficient evidence that the mean sugar content differs from 12 g.

C4 Applies the general procedure for testing a hypothesis with fixed probability in context situations.

Problem 1 — Online Quiz Completion Rate

Problem. An online course expects at least 80 % of students to complete weekly quizzes. In a week with $n = 150$ students, 114 complete the quiz.

- Population parameter: completion proportion p .
- Significance level: $\alpha = 0,05$.

Question. Apply the general hypothesis-testing procedure to decide if the completion rate is below 80 %.

Solution.

Step 1: Hypotheses.

$$H_0 : p = 0,80$$

$$H_a : p < 0,80$$

Step 2: Type of test. One-tailed (left), because we test for a drop below 80 %.

Step 3: Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,76 - 0,80}{\sqrt{0,80(0,20)/150}} \approx \frac{-0,04}{0,0326} \approx -1,23.$$

Step 4: Reference distribution. Standard normal under H_0 .

Step 5: P-value. $P(Z \leq -1,23) \approx 0,1093$.

Step 6: Decision. Since $0,1093 > 0,05$, fail to reject H_0 .

Conclusion. The data do not show that completion is below 80 %.

Problem 2 — School ID Card Errors

Problem. The administration wants the ID card printing error rate to be at most 2 %. In a sample of $n = 500$ cards, 14 have errors.

- Population parameter: error proportion p .
- Significance level: $\alpha = 0,01$.

Question. Use the general testing procedure to decide whether the error rate exceeds 2 %.

Solution.

Step 1: Hypotheses.

$$H_0 : p = 0,02$$

$$H_a : p > 0,02$$

Step 2: Type of test. One-tailed (right), because we test for an increase above 2 %.

Step 3: Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,028 - 0,02}{\sqrt{0,02(0,98)/500}} \approx \frac{0,008}{0,0063} \approx 1,27.$$

Step 4: Reference distribution. $Z \sim N(0, 1)$.

Step 5: P-value. $P(Z \geq 1,27) \approx 0,1020$.

Step 6: Decision. Since $0,1020 > 0,01$, fail to reject H_0 .

Conclusion. There is not enough evidence that the error rate is above 2 %.

C5 Calculates hypothesis tests on the mean of a population.

Problem 1 — Seed Germination Height

Problem. A biology class believes a fertilizer increases the mean seedling height beyond 18 cm. A sample of $n = 16$ seedlings grown with the fertilizer has mean $\bar{x} = 19,2$ cm. Assume $\sigma = 2,4$ cm.

- Population parameter: mean height μ (cm).
- Significance level: $\alpha = 0,05$.

Question. Test whether the fertilizer raises the mean height.

Solution.

Hypotheses.

$$H_0 : \mu = 18$$

$$H_a : \mu > 18$$

Type of test. One-tailed (right), because we test for an increase.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{19,2 - 18}{2,4/\sqrt{16}} = \frac{1,2}{0,6} = 2,00.$$

Reference distribution. Standard normal under H_0 .

P-value. $P(Z \geq 2,00) = 0,0228$.

Decision. Since $0,0228 < 0,05$, reject H_0 .

Conclusion. The data suggest the fertilizer increases mean seedling height.

Problem 2 — Printer Ink Usage

Problem. A school printer is expected to use 150 mL of ink per week. Over $n = 20$ weeks, the sample mean is $\bar{x} = 156$ mL with $\sigma = 18$ mL.

- Population parameter: mean weekly ink use μ (mL).
- Significance level: $\alpha = 0,05$.

Question. Determine if the mean ink usage differs from 150 mL.

Solution.

Hypotheses.

$$H_0 : \mu = 150$$

$$H_a : \mu \neq 150$$

Type of test. Two-tailed because any deviation from 150 mL matters.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{156 - 150}{18/\sqrt{20}} \approx \frac{6}{4,0249} \approx 1,49.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $2P(Z \geq 1,49) \approx 2(0,0681) = 0,1362$.

Decision. Since $0,1362 > 0,05$, fail to reject H_0 .

Conclusion. There is insufficient evidence that mean ink usage differs from 150 mL.

C6 Establishes the correct hypothesis for a test about a population proportion.

Problem 1 — Recycling Participation

Problem. The school hopes that at least 60 % of students participate in the recycling program. In a survey of $n = 120$ students, 66 report participating.

- Population parameter: participation proportion p .
- Significance level: $\alpha = 0,05$.

Question. Establish the correct hypotheses and conduct the test.

Solution.

Hypotheses.

$$H_0 : p = 0,60$$

$$H_a : p < 0,60$$

Type of test. One-tailed (left), because we test if participation is below 60 %.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,55 - 0,60}{\sqrt{0,60(0,40)/120}} \approx \frac{-0,05}{0,0447} \approx -1,12.$$

Reference distribution. Standard normal under H_0 .

P-value. $P(Z \leq -1,12) \approx 0,1314$.

Decision. Since $0,1314 > 0,05$, fail to reject H_0 .

Conclusion. The survey does not provide enough evidence that participation is below 60 %.

Problem 2 — Mobile Payment Adoption

Problem. A city report suggests that 40 % of small shops use mobile payments. A local business group believes the adoption rate is higher. In a sample of $n = 180$ shops, 90 use mobile payments.

- Population parameter: adoption proportion p .
- Significance level: $\alpha = 0,05$.

Question. Establish the correct hypotheses and perform the test.

Solution.

Hypotheses.

$$H_0 : p = 0,40$$

$$H_a : p > 0,40$$

Type of test. One-tailed (right), because the research question asks if adoption is higher.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,50 - 0,40}{\sqrt{0,40(0,60)/180}} \approx \frac{0,10}{0,0365} \approx 2,74.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $P(Z \geq 2,74) \approx 0,0031$.

Decision. Since $0,0031 < 0,05$, reject H_0 .

Conclusion. There is evidence that mobile payment adoption exceeds 40 %.

C7 Contrasts P-values in hypothesis tests with the reference probability.

Problem 1 — Tablet Screen Defects

Problem. A manufacturer claims that only 3 % of tablets have screen defects. In a shipment of $n = 400$ tablets, 18 are defective.

- Population parameter: defect proportion p .
- Significance level: $\alpha = 0,05$.

Question. Compute the P-value and compare it with the reference probability to decide whether the defect rate is higher than claimed.

Solution.

Hypotheses.

$$H_0 : p = 0,03$$

$$H_a : p > 0,03$$

Type of test. One-tailed (right) because the concern is a higher defect rate.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,045 - 0,03}{\sqrt{0,03(0,97)/400}} \approx \frac{0,015}{0,0085} \approx 1,76.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $P(Z \geq 1,76) \approx 0,0392$.

Decision. Since $0,0392 < 0,05$, reject H_0 .

Conclusion. The P-value is smaller than the reference probability $\alpha = 0,05$, so the data indicate a higher defect rate.

Problem 2 — Volunteer Attendance

Problem. A community center expects that 70 % of registered volunteers attend monthly meetings. In a sample of $n = 90$ volunteers, 60 attend.

- Population parameter: attendance proportion p .
- Significance level: $\alpha = 0,10$.

Question. Calculate the P-value and compare it with α to decide if attendance is below 70 %.

Solution.

Hypotheses.

$$H_0 : p = 0,70$$

$$H_a : p < 0,70$$

Type of test. One-tailed (left), because we test for a decrease in attendance.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,667 - 0,70}{\sqrt{0,70(0,30)/90}} \approx \frac{-0,033}{0,0483} \approx -0,68.$$

Reference distribution. Standard normal under H_0 .

P-value. $P(Z \leq -0,68) \approx 0,2483$.

Decision. Since $0,2483 > 0,10$, fail to reject H_0 .

Conclusion. The P-value exceeds the reference probability, so there is not enough evidence that attendance is below 70 %.

C8 Analyses the relationship between hypothesis testing and confidence interval estimation.

Problem 1 — Classroom Noise Level

Problem. The school claims that the mean noise level during group work is 65 dB. A random sample of $n = 36$ readings gives $\bar{x} = 66,5$ dB with $\sigma = 6$ dB.

- Population parameter: mean noise level μ (dB).
- Significance level: $\alpha = 0,05$.

Question. Use a hypothesis test and a confidence interval to evaluate the claim and explain the link between them.

Solution.

Hypotheses.

$$H_0 : \mu = 65$$

$$H_a : \mu \neq 65$$

Type of test. Two-tailed because any difference from 65 dB is important.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{66,5 - 65}{6/\sqrt{36}} = \frac{1,5}{1} = 1,50.$$

P-value. $2P(Z \geq 1,50) \approx 2(0,0668) = 0,1336$.

Decision. Since $0,1336 > 0,05$, fail to reject H_0 .

Confidence interval link. A 95 % confidence interval is

$$\bar{x} \pm z_{0,025} \frac{\sigma}{\sqrt{n}} = 66,5 \pm 1,96(1) = (64,54, 68,46).$$

Because 65 dB lies in the interval, the two-tailed test at $\alpha = 0,05$ does not reject H_0 .

Conclusion. Both methods suggest the data do not show a significant difference from 65 dB.

Problem 2 — School Lunch Satisfaction

Problem. The school cafeteria reports that 55 % of students are satisfied with lunch quality. A survey of $n = 200$ students finds 120 are satisfied.

- Population parameter: satisfaction proportion p .
- Significance level: $\alpha = 0,10$.

Question. Use a hypothesis test and a confidence interval to analyze the claim and explain their connection.

Solution.

Hypotheses.

$$H_0 : p = 0,55$$

$$H_a : p \neq 0,55$$

Type of test. Two-tailed because deviations in either direction matter.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,60 - 0,55}{\sqrt{0,55(0,45)/200}} \approx \frac{0,05}{0,0352} \approx 1,42.$$

P-value. $2P(Z \geq 1,42) \approx 2(0,0778) = 0,1556$.

Decision. Since $0,1556 > 0,10$, fail to reject H_0 .

Confidence interval link. A 90 % confidence interval for p is

$$\hat{p} \pm z_{0,05} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 0,60 \pm 1,645 \sqrt{\frac{0,60(0,40)}{200}} \approx 0,60 \pm 0,057.$$

So the interval is approximately $(0,543, 0,657)$. Because 0,55 is inside this interval, the two-tailed test at $\alpha = 0,10$ does not reject H_0 .

Conclusion. The confidence interval and test agree that the data do not show a significant difference from 55 %.

C9 Evaluates finance and economic context situations in which a statistical hypothesis appears.

Problem 1 — Microloan Default Rate

Problem. A community bank expects that at most 4 % of microloans default. This affects interest pricing and reserves. In the last year, $n = 500$ loans were issued and 28 defaulted.

- Population parameter: default proportion p .
- Significance level: $\alpha = 0,05$.

Question. Test whether the default rate exceeds 4% and explain why a hypothesis test is needed.

Solution.

A hypothesis test is needed because the bank must decide if the observed defaults suggest a real increase in the population default rate, which would require higher reserves.

Hypotheses.

$$H_0 : p = 0,04$$

$$H_a : p > 0,04$$

Type of test. One-tailed (right), because we test for a higher default rate.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,056 - 0,04}{\sqrt{0,04(0,96)/500}} \approx \frac{0,016}{0,0088} \approx 1,82.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $P(Z \geq 1,82) \approx 0,0344$.

Decision. Since $0,0344 < 0,05$, reject H_0 .

Conclusion. The data suggest the default rate is above 4%, so the bank should consider higher reserves.

Problem 2 — Market Stall Daily Revenue

Problem. A cooperative of market vendors estimates that the mean daily revenue per stall is \$120. To decide whether a new layout improved sales, they collect $n = 25$ days of data with mean $\bar{x} = 126$ and known $\sigma = 15$.

- Population parameter: mean revenue μ (USD).
- Significance level: $\alpha = 0,10$.

Question. Test whether the mean revenue has increased, and explain the economic decision involved.

Solution.

A hypothesis test is required because the cooperative must decide if the observed increase is large enough to justify keeping the new layout.

Hypotheses.

$$H_0 : \mu = 120$$

$$H_a : \mu > 120$$

Type of test. One-tailed (right), because the decision depends on improvement.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{126 - 120}{15/\sqrt{25}} = \frac{6}{3} = 2,00.$$

Reference distribution. Standard normal under H_0 .

P-value. $P(Z \geq 2,00) = 0,0228$.

Decision. Since $0,0228 < 0,10$, reject H_0 .

Conclusion. The data provide evidence that mean daily revenue increased, supporting the new layout.

C10 Models a research question in which a statistical hypothesis must be tested.

Problem 1 — Savings Program Impact

Problem. A school introduces a savings club and wants to know if the average monthly savings of participants exceeds \$15. In a sample of $n = 30$ students, the mean savings is $\bar{x} = 16,8$ with $\sigma = 4,2$.

- Population parameter: mean monthly savings μ (USD).
- Significance level: $\alpha = 0,05$.

Question. Model the research question with a statistical hypothesis test and provide the decision.

Solution.

The research question “Does the savings club increase mean monthly savings above \$15?” is modeled by testing whether μ exceeds 15.

Hypotheses.

$$H_0 : \mu = 15$$

$$H_a : \mu > 15$$

Type of test. One-tailed (right), because we test for an increase.

Test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{16,8 - 15}{4,2/\sqrt{30}} \approx \frac{1,8}{0,767} \approx 2,35.$$

Reference distribution. $Z \sim N(0, 1)$.

P-value. $P(Z \geq 2,35) \approx 0,0094$.

Decision. Since $0,0094 < 0,05$, reject H_0 .

Conclusion. The data support that the savings club increases average monthly savings beyond \$15.

Problem 2 — Student-led Coffee Cart Sales

Problem. A student entrepreneurship group wants to test whether the proportion of customers who return within a week is at least 50 % after introducing a loyalty stamp card. In $n = 120$ customers, 54 return within a week.

- Population parameter: return proportion p .
- Significance level: $\alpha = 0,05$.

Question. Model the research question with a hypothesis test and state the conclusion.

Solution.

The research question is whether the return rate reaches the target of 50 %, so we test if p is below 0,50.

Hypotheses.

$$H_0 : p = 0,50$$

$$H_a : p < 0,50$$

Type of test. One-tailed (left), because the concern is not meeting the target rate.

Test statistic.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0,45 - 0,50}{\sqrt{0,50(0,50)/120}} \approx \frac{-0,05}{0,0456} \approx -1,10.$$

Reference distribution. Standard normal under H_0 .

P-value. $P(Z \leq -1,10) \approx 0,1357$.

Decision. Since $0,1357 > 0,05$, fail to reject H_0 .

Conclusion. The data do not show the return rate is below 50 %, so the target cannot be ruled out.